MIL-I-8691A (ASG)
17 OCTOBER 1955
Superseding
MIL-I-8691(Aer)
15 December 1954

MILITARY SPECIFICATION

INDICATOR, AZIMUTH; ID-307/ARN

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 Scope.- This specification covers one type of airborne instrument, designated ID-307/ARN, for use with a receiver-transmitter for presentation of the bearing of an aircraft from a navigation beacon.

1.2 Classification.- The instrument shall be a unit of the following type:

Major unit	Type designation	Size	Weight
Azimuth indicator	ID-307/ARN	3-1/4 in. sq by 6-5/8 in. lg	3.0 lb

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-I-7028	Instruments and Instrument Boards; Aircraft, Installation of
MIL-L-25142	Luminescent Material, Fluorescent
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MS33550	Case, Instrument, 2-1/2 Dial, Without Sump, Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN3057	Adapter, Electrical Accessory to Cable
AN3102	Connectors - Electrical, Receptacles, Box Mounting
AN3108	Connectors - Electrical, Plugs, Angle 90°

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning quantity production, a preproduction sample representative of the production item shall be submitted to determine compliance with all the requirements of this specification.

3.2 General.- This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.3 Parts and materials.- In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(a) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(b) When the contractors can demonstrate that the use of standard MS, AN, or JAN parts or materials will not fulfill the design objectives because of size, weight, performance, or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(c) When a nonstandard part is used, the meeting of general material requirements, such as noninflammability, fungus resistance, nontoxicity, etc. and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-I-5422, shall be of prime consideration.

(d) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be so constructed that the unit as a whole will pass material, electrical, and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the procuring activity.

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3.3.1 Nonstandard part approval and material approval.- Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows:

(a) Mechanical parts are expected to meet the material requirements of Specification MIL-E-5400, and shall be submitted as nonstandard parts only when the material requirements are not met.

(b) Nonstandard parts, as defined in Specification MIL-E-5400, of an electrical nature must be approved before the preproduction model of the equipment is delivered to the procuring activity. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the "Nonstandard Part Data Sheet," the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.3.2 Materials used to attain design objectives.- Minimum size and weight, simplicity of operation, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts, and processes other than those required by Specification MIL-E-5400 shall be investigated, and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their use, a request for approval of a nonstandard part shall be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

3.3.3 Corrosion resistance.- Materials and parts shall be of corrosion-resisting types or shall be suitably processed or finished in accordance with type I finishes of Specification MIL-F-14072 to meet the specified service conditions.

3.3.4 Fungus resistance.- Materials and parts shall be of the fungus-resisting types or shall be suitably processed or finished to meet the specified service conditio....

3.3.5 Nonmagnetic materials.- Nonmagnetic materials shall be used in the instrument to as great an extent as possible where magnetic materials are not essential to meet the design requirements.

3.4 Design and construction.-

3.4.1 Azimuth indicator ID-307/ARN.- The azimuth indicator shall be designed to receive power and control signals from an RT-220/ARN-21-type receiver-transmitter and a C-2-type compass or compass adapter to actuate a pointer to indicate bearing of the receiver-transmitter from a navigation beacon, and provide one output for feeding a CV-279/ARN-type phase detecting network and another output for supplying more accurate (vernier) bearing information.

3.4.2 Indicator housing.- The indicator shall be completely housed in a 3-1/8 inch diameter case similar to that illustrated in Standard MS33550 with a bezel, cover glass, and connector, and shall not exceed the outline dimensions shown on figure 1. The case, bezel, cover glass, and connector shall be designed to fit together with the indicator frame to form a dustproof, dirtproof housing for the indicator.

3.4.2.1 Case.- The case shall be made from aluminum alloy suitably treated and finished in durable dull black. The case shall be secured to the indicator frame with screws in such manner that it may be removed without disturbing the connector or other parts.

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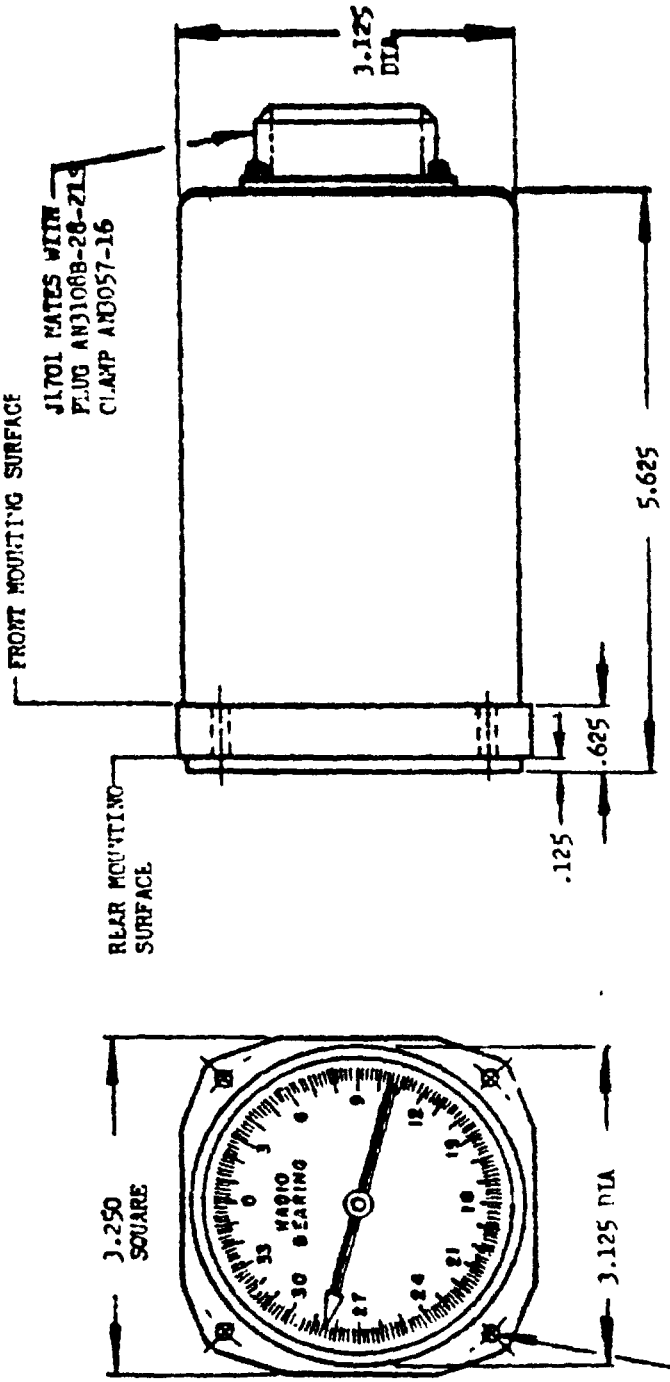


FIGURE 1. ID-307/ARN INDICATOR, AZIMUTH, OUTLINE DIMENSIONS

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3.4.2.2 Bezel.- The bezel shall be made from non-ferrous low-density metal suitably treated and finished in durable dull black. The bezel shall be neatly fitted and secured to the indicator frame with screws. The bezel shall retain the cover glass securely when assembled but shall be so designed that the glass is readily replaceable when necessary.

3.4.2.3 Cover glass.- The cover glass shall be clear, flat, and free from flaws which would interfere with normal reading of the indicator.

3.4.3 Mounting.- The indicator shall be designed for both front and rear mounting with machine screws in accordance with Specification MIL-I-7028.

3.4.4 Components.- The indicator shall be designed with the following basic component parts connected as shown on figure 2.

3.4.4.1 Motor-rate generator.- A two-phase induction motor-rate generator with a motor rating of 18 volts, 400 cps, per phase and a generator rating of 0.3 volt a-c output per 1,000 revolutions per minute shall be designed to drive the indicator pointer, potentiometer, differential transmitter, and resolvers through associated gearing and supply an a-c voltage proportional to the speed of rotation. The motor-rate generator shall be similar and equal to Eclipse-Pioneer type FV-101-2-A1.

3.4.4.2 Bearing measuring potentiometer.- A precision sinusoidal potentiometer rated 150 volts, 16,000 ohms ±10 percent, angular accuracy ±0.6 degree, amplitude accuracy ±0.8 percent with 360 degrees of rotation shall be designed to determine the phase difference between two 15 cps signals for measuring bearing in conjunction with the bearing measuring resolver. The bearing measuring potentiometer shall be electrically equal to Ganewell type RL-11C.

3.4.4.3 Bearing measuring resolver.- A miniature precision resolver with a single-phase rotor rated at 3 volts, 135 cps with 0.010 volt d-c input and a two-phase stator rated at 9 volts 135-cps maximum coupling output shall be designed to determine the phase difference between two 135-cps signals for measuring bearing in conjunction with the bearing measuring potentiometer. The d-c resistance of the resolver rotor shall not exceed 60 ohms. The impedance of the resolver rotor shall exceed 90 ohms at 135 cps. The total harmonic distortion of a pure 135-cps sine wave shall not exceed 2 percent. The bearing measuring resolver shall be similar and equal to Eclipse-Pioneer type AY-163-A1.

3.4.4.4 Differential synchro transmitter.- A synchro transmitter of the differential control type rated at 26 volts, 400 cps, shall be designed to produce a signal with a phase equal to the phase difference between the compass heading signal and the receiver-transmitter bearing signal. The synchro transmitter shall be similar and equal to Eclipse-Pioneer type AY-23ZS-4-B.

3.4.4.5 Fundamental synchro resolver.- A synchro resolver rated at 26 volts, 400 cps, with a two-phase rotor and a two-phase stator shall be designed to operate at the rate and precision of the pointer shaft to produce an output directly related to the bearing indication. The synchro resolver shall be similar and equal to Eclipse-Pioneer type AY-221S-3-B.

3.4.4.6 Harmonic synchro resolver.- A miniature synchro resolver rated at 26 volts, 400 cps, with a two-phase rotor and a two-phase stator shall be designed to operate at the rate and precision of the bearing measuring resolver shaft to produce an output directly related to the ninth harmonic of the fundamental bearing signal. The miniature synchro resolver shall be similar and equal to Eclipse-Pioneer type AY-523-3-A1.

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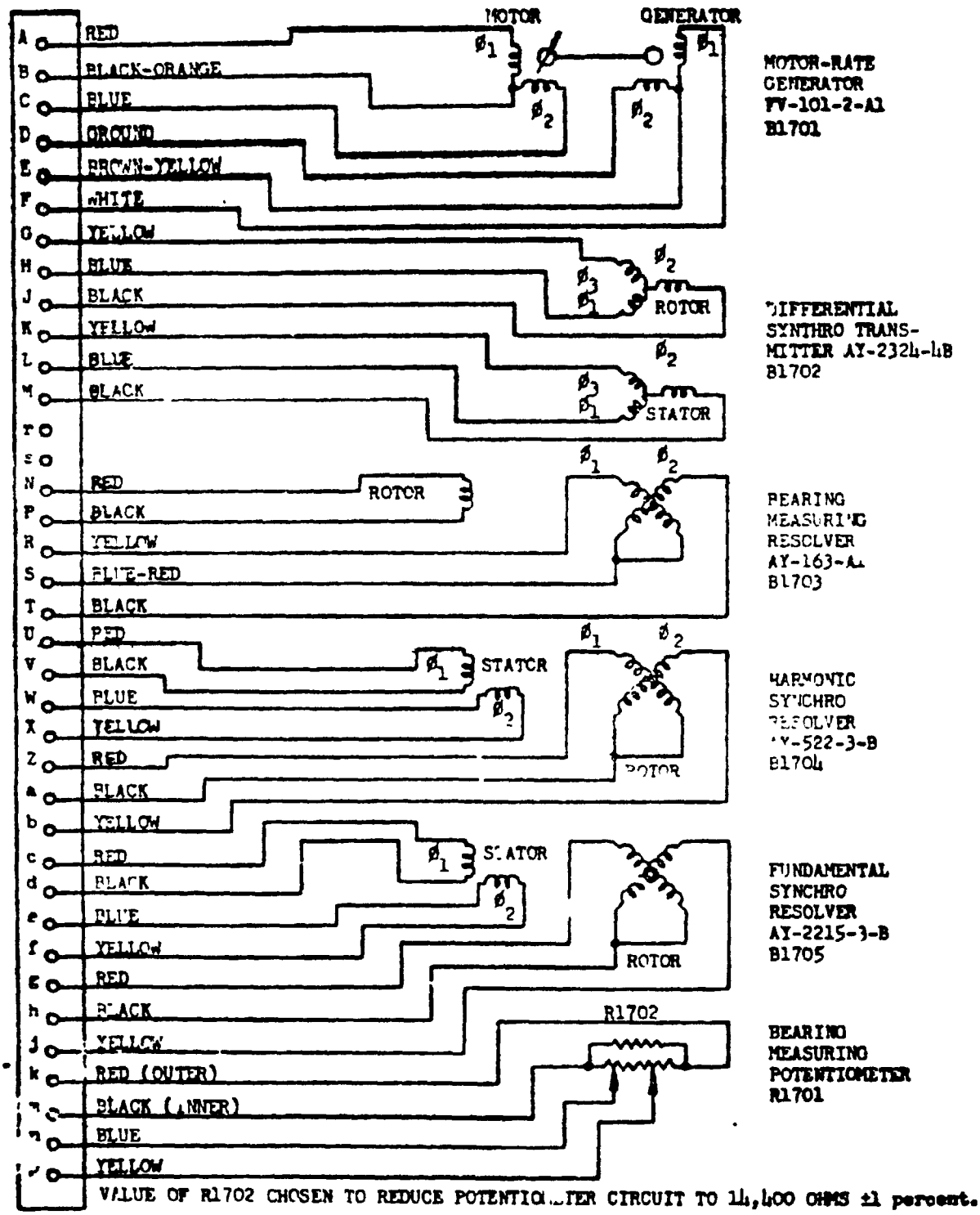


FIGURE 2. Schematic, ID-307/ARN; indicator, azimuth

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3.4.4.7 Dial.- The indicator shall be designed with a 2-3/4 inch fixed dial with the 0 degree mark oriented in the 12 o'clock position. The circumference of the dial shall be divided into 36 major divisions and marked with heavy lines. Every third division shall be marked with numerals to indicate one tenth the number of degrees the mark represents. Four lighter and shorter lines shall be marked between each major division to provide a 2-degree scale. The lettering "RADIO" and "BEARING" in 1/8-inch-high characters shall be centered on the dial. The lettering and numbering shall conform to Standard MS33558. The lettering, numbering, and major division degree marks shall be coated with fluorescent-luminescent material in accordance with Specification MIL-L-25142. The background of the dial shall be finished in durable dull black. Any other lettering appearing on the dial shall be black on black.

3.4.4.8 Pointer.- The pointer shall be a thin, single-bar type with an arrow-shaped head and a wedge-shaped tail. The pointer shall be of sufficient length to extend over one quarter of the 2-degree scale marks at both the head and tail when the pointer is mounted in the indicator. The pointer shall be marked with fluorescent-luminescent material in accordance with Specification MIL-L-25142. The background of the pointer shall be durable dull black. The pointer shall be mounted as close to the dial face as practicable in order to reduce parallax to a minimum.

3.4.4.9 Connector.- An AN3102C-28-21P connector in accordance with Specification MIL-C-5015 shall be mounted on the rear of the instrument to effect all electrical connections. The pin connections shall be made as specified in table I.

TABLE I

Pin connections

Component			Connection	Pin
Motor-rate generator (FV-101-2-A1)	Motor		Phase 1 (fixed) stator	A
			Common stator	B
			Phase 2 (control) stator	C
	Rate Generator	High Z output	Phase 2 stator	D
			Phase 2 stator	E
		Low Z excitation	Phase 1 stator	E
			Phase 1 stator	F
Differential synchro transmitter (AY-232S-4-B)			Rotor	G
		Indicator output	Rotor	H
			Rotor	J
			Stator	K
		Compass input	Stator	L
			Stator	M
Bearing measuring resolver (AY-163-A1)		Input	Rotor	N
			Rotor	P
		Output	Phase 1 stator	R
			Common stator	S
			Phase 2 stator	T
Harmonic synchro resolver (AY-523-3-A1)			Phase 1 stator	U
			Phase 1 stator	V
			Phase 2 stator	W
			Phase 2 stator	X
			Phase 1 rotor	Z
			Common rotor	a
			Phase 2 rotor	b

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TABLE I (continued)

Pin connections

Component		Connection	Pin
Fundamental synchro resolver (AT-221S-3-B)		Phase 1 stator	c
		Phase 1 stator	d
		Phase 2 stator	e
		Phase 2 stator	f
		Phase 1 rotor	g
		Common rotor	h
		Phase 2 rotor	j
Bearing measuring potentiometer	Input	Card	k
		Card	m
	Output	Contract arm 1	n
		Contract arm 2	p
		No connection	r
		No connection	s

3.4.4.10 Adjusting screw.- An external adjusting screw shall be designed to set the zero of the fundamental synchro resolver over a ±5-degree range.

3.4.5 Gearing and rotation.- The pointer shall be driven by the motor in a clockwise direction for increasing bearing when the voltage applied to the control phase, (phase two) of the motor lags the voltage applied to the fixed phase (phase one). The bearing measuring resolver and the harmonic synchro resolver shall rotate at a rate equal to nine times that of the pointer, bearing measuring potentiometer, differential synchro transmitter, or fundamental synchro resolver.

3.4.6 Backlash.- The angular backlash between the pointer and the motor pinion shall not exceed 40 degrees as measured at the motor pinion. The backlash slack measured at the pointer shall not exceed 0.2 degree total.

3.4.7 Visibility.- The bezel, dial, and pointer shall be so designed that when they are assembled the bearing display will be visible and readable within a 60-degree cone about the axis of the indicator. The pointer and the dial shall be positioned to keep the parallax to a practical minimum.

3.4.8 Interchangeability.- Mechanical and electrical interchangeability shall exist between like assemblies, subassemblies, and replaceable parts, regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not mean identity but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the equipment or assemblies, including cabling, wiring, and mounting and without resorting to selection. In the design of the equipment, provisions shall be made for design tolerances sufficient to accommodate various sizes and characteristics of any one type of article having the limiting dimensions and characteristics set forth in the specifications for the particular component involved, without departure from the specified performance.

3.4.9 Service conditions.-

3.4.9.1 Electrical.- Electrical service conditions shall be used for reference and design purposes. Testing shall be done in accordance with Section 4.

3.4.9.1.1 General.- The main power and control signals for the indicator shall be supplied from an AT-220/ARN-21 radio receiver-transmitter. Control signals for the differential synchro transmitter shall be supplied by the J-2 compass adapter.

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3.4.9.1.2 Motor-rate generator input.- The motor portion of the motor-rate generator shall be designed to operate with 18 volts ±10 percent, 380 to 420 cps continuously applied to the fixed phase and 0 to 18 volts, 380 to 420 cps with a variable phase from 90 degrees lagging to 90 degrees leading applied to the control phase of the motor. The motor-rate generator shall be designed to operate with 18 volts ±10 percent, 380 to 420 cps continuously applied to phase one.

3.4.9.1.3 Bearing measuring potentiometer input.- The bearing potentiometer circuit shall be designed to maintain specified performance with 15 volts, 15 cps applied to the card continuously.

3.4.9.1.4 Bearing measuring resolver input.- The bearing measuring resolver circuit shall be designed to maintain specified performance with 0.7 volt, 135 cps applied to the rotor continuously.

3.4.9.1.5 Resolver input.- The fundamental and harmonic synchro resolver will be supplied with 18 volts, 380 to 420 cps for continuous operation.

3.4.9.2 Environmental.- Environmental service conditions shall be used for reference and design purposes. Testing shall be done in accordance with Section 4.

3.4.9.2.1 General.- The indicators built to this specification will be subjected to the severe climatic conditions encountered in the operation of military aircraft, and shall therefore be so designed that no fixed parts become loose, no moveable parts become undesirably sluggish or excessively free, and that there will be as little performance degradation or life deterioration as possible under specified environmental service conditions.

3.4.9.2.2 Operating conditions.- The indicators shall be designed to perform under any of the following or any natural combination of the following conditions:

(a) Continuous operation over an ambient temperature range from -55°C to +55°C with the temperature remaining constant for long periods or varying at a rate up to 1°C per second.

(b) Continuous operation over a barometric pressure range from 30 in. Hg to 1.32 in. Hg with the pressure remaining constant for long periods or varying at a rate up to 0.5 in. Hg per second.

(c) Short period operation (30 minutes or less) at a temperature of 71°C with a barometric pressure of 30 in. Hg.

(d) Continuous and intermittent operation at relative humidities up to 100 percent with temperatures up to 55°C, including conditions where condensation takes place in and on the instrument.

(e) Intermittent operation in salt spray and salt-laden atmosphere.

(f) Intermittent operation in atmosphere filled with explosive vapors.

(g) Intermittent operation in fine dust-laden atmosphere.

(h) Intermittent operation in atmosphere laden with fungus spores.

3.4.9.2.3 Nonoperating conditions.- The indicators shall be designed to recover from any of the following or any natural combination of the following conditions:

(a) Inoperative periods at temperatures ranging from -65°C to +85°C with the temperature remaining constant for long periods or varying at a rate up to 1°C per second.

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(b) Inoperative periods at barometric pressure ranging from 30 in. Hg to 1.32 in. Hg with the pressure remaining constant for long periods or varying at a rate up to 0.5 in. Hg per second.

(c) Inoperative periods at relative humidities up to 100 percent with temperatures up to 55°C, including conditions where condensation takes place in and on the instrument.

(d) Inoperative periods in salt spray and salt-laden atmosphere.

(e) Inoperative periods in atmosphere filled with explosive vapors.

(f) Inoperative periods in fine dust-laden atmosphere.

(g) Inoperative periods in atmosphere laden with fungus spores.

3.4.9.3 Mechanical.- Mechanical service conditions shall be used for reference and design purposes. Testing shall be done in accordance with Section 4.

3.4.9.3.1 General.- The indicators built to this specification will be subjected to the severe physical conditions encountered in field handling and servicing and in operation of military aircraft, and shall therefore be so designed that no fixed parts become loose, no moveable parts become unadjusted, and that there will be as little performance degradation or life deterioration as possible under specified mechanical service conditions.

3.4.9.3.2 Operating and nonoperation conditions.- When the indicator is mounted on an aircraft-type instrument board in accordance with Specification MIL-P-7029 or packed in accordance with the recommendations of the manufacturer, it shall be designed to perform under or recover from any of the following conditions:

(a) Continuous operation and recover from inoperative periods under vibration having a total excursion of 0.06 inch at any frequency within the range from 10 to 55 cps.

(b) During operative or inoperative periods, sustain and recover from shocks up to 30g with time durations of 11 ±1 milliseconds.

(c) Continuous operation and recovery from long inoperative periods when oriented in any position.

3.5 Screw threads.- Screw threads shall be in accordance with Specification MIL-S-7742.

3.6 Weight.- The weight of the complete indicator shall not exceed 3 pounds.

3.7 Performance.-

3.7.1 Operation.- The azimuth indicator, when properly interconnected to an RT-220/ARN-21-type receiver-transmitter and associated components supplying the required signal and control voltages, shall operate to perform the following functions:

(a) Automatically and continuously indicate by means of a pointer against a fixed dial the bearing of a navigation beacon from the receiver-transmitter.

(b) With an input from a G-2-type compass or compass adaptor, provide an output to operate the pointer of ID-250A/ARN-type course indicator to indicate the relative bearing of a navigation beacon from the receiver-transmitter.

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(c) Provide a fundamental synchro resolver output which will supply a navigation computer with bearing information, and which when combined with the output of the course selector synchro resolver of an ID-249A/ARN-, ID-351A/ARN-, or ID-387/ARN-type course indicator, as designated by the procuring activity, in a CV-279/ARN-type phase detector may be utilized to show deviation from a selected course on the vertical pointer of the course indicator and to resolve the course ambiguity by operating the "To-From" flag of the course indicator.

(d) Provide a vernier bearing output signal from a harmonic synchro resolver operating at nine times the rate of the fundamental synchro resolver.

3.7.2 Accuracy.- The accuracy of the indicators shall fall within the following limits and shall not exceed these limits, except as specified in Section 4.

3.7.2.1 Indicated bearing accuracy.- The accuracy of the pointer as read against the dial from 0 through 360 degrees shall be within ±0.5 degree of the bearing measured by a receiver-transmitter (or the bearing represented by a position test signal).

3.7.2.2 Relative bearing output accuracy.- The accuracy of the relative bearing output shall be within ±0.1 degree of the difference between the bearing measured by the receiver-transmitter (or the bearing represented by a position test signal) and heading determined by a compass (or the heading represented by a position test signal) from 0 through 360 degrees with 0 output occurring at 180 degrees.

3.7.2.3 Fundamental synchro resolver output accuracy.- The output of the fundamental synchro resolver shall be within ±0.5 degree of the bearing measured by a receiver-transmitter (or the bearing represented by a position test signal) from 0 through 360 degrees with 0 output occurring at 345 degrees adjustable ±5 degrees.

3.7.2.4 Harmonic synchro resolver output accuracy.- The output of the harmonic synchro resolver shall be within ±0.8 degree of the bearing measured by a receiver-transmitter (or the bearing represented by a position test signal) from 0 through 360 degrees with 0 output occurring at 0 degree.

3.7.2.5 Dial marking accuracy.- The degree scale markings on the dial shall be on the correct bearing angles to within ±0.25 degree.

3.7.3 Sensitivity.- The sensitivity of the indicator shall be such that with the application of 18 volts, 400 cps to the fixed phase of the motor, the application of 1 volt, 400 cps leading or lagging 90 degrees to the control phase will cause the motor to start with the pointer in any position and continue to rotate smoothly.

3.7.4 Indicator speed.- With 18 volts, 400 cps applied to both phases of the motor and phase one of the rate generator, and with the voltage across the control phase leading or lagging the voltage across the fixed phase by 90 degrees, the pointer shall be driven backward or forward at a rate equal to 24 degrees per second ±12 percent.

3.7.5 Service period.- The instruments shall have a reliable operating period of at least 500 hours without requiring removal for bench servicing. Parts requiring servicing or replacement within this interval to renew this service period shall be specified by the manufacturer.

3.7.6 Total life.- The instruments shall have a minimum total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring servicing or replacement within this period and the life of such parts shall be specified by the manufacturer.

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3.8 Identification of product.-

3.8.1 Nomenclature and nameplates.- Nomenclature assignment and nameplate approval shall be in accordance with the requirements of Specification MIL-N-18307.

3.8.2 Patent information.- Patent numbers, patent license notices, or any reference to either shall not appear on a nameplate. If the contractor considers it necessary to apply such a patent notice initially to the equipment because of some contractual obligation to a licensor or because of the contractor's interpretation of the patent statutes, the notice may be applied only under the following conditions for secret and confidential articles of equipment:

(a) The notice shall be supplied as part of a tag, label, or sticker which may be easily removed or torn off by the Government Inspector.

(b) The appearance and location of the tag, label, or sticker shall be such as to render unlikely the overlooking of its removal by the Government Inspector at time of acceptance inspection.

(c) The contractor shall institute a checkup as part of the procedure of packing for shipment to assure that the notice has been removed and shall point out to the Government Inspector any articles from which the notice has not been removed.

(d) The tag, label, or sticker shall not bear the Government-type designation of the article of equipment to which it applies.

3.9 Workmanship.- Workmanship, fabrication, and manufacturing processes shall conform to the requirements of Specification MIL-E-5400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of aircraft electronic equipment shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed on samples representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification. Preproduction tests shall consist of the following two tests:

(1) Contractor demonstration tests - Contractor demonstration tests are accomplished under the responsibility of the contractor on a sample representative of the production equipment, to demonstrate that the equipment complies with all the requirements of the applicable specifications.

(2) Service approval tests - Service approval tests are accomplished under the responsibility of the procuring activity on a sample representative of the production equipment, to determine that the equipment meets all the requirements of the specification. These tests may be accomplished by having the procuring activity supervise the Contractor demonstration tests.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance under contract.

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4.2 Preproduction tests.-

4.2.1 Contractor demonstration tests.- The contractor shall conduct tests on one or more sample equipments as necessary to determine that the design of the equipment as proposed by the contractor will meet the requirements of all applicable specifications. Contractor demonstration tests shall be conducted in accordance with the approved Preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the Service approval test sample. The Inspector and the procuring activity shall be advised when tests are to be conducted, in order that a procuring activity representative may witness or supervise the test when so desired. When tests are supervised, the procuring activity may decide to omit certain portions of the Service approval tests which will duplicate the Contractor demonstration tests; however, in conducting Service approval tests the procuring activity may elect to repeat any test previously conducted by the contractor, when deemed necessary. Contractors not having laboratory facilities to satisfactorily conduct all tests shall obtain either the services of a commercial testing laboratory or written approval from the procuring activity to omit that portion of the tests which cannot be conducted.

4.2.1.1 Examination of instrument.- The indicator shall be examined visually and mechanically as necessary to determine that the instrument fully complies with the physical requirements of this specification in regard to materials, dimensions, workmanship, marking, and finish.

4.2.1.2 Electrical tests.- Electrical tests shall be conducted under standard conditions unless otherwise noted.

4.2.1.2.1 Rotation speed.- With 18 volts, 400 cps applied to phase one of the motor and rate generator and 18 volts, 400 cps first lagging by 90 degrees and then leading by 90 degrees applied to the control phase (phase two) of the motor, the time required for the pointer to turn through 1,440° ±12 percent shall be equal to 60 seconds for either direction of rotation. The time may be measured with a laboratory-type stopwatch with a rotary second hand accurate to 0.1 second.

4.2.1.2.2 Rate-generator output.- With the indicator connected as specified in 4.2.1.2.1, the speed when adjusted for a rate-generator output of 2 volts shall be 1,080 degrees per minute ±12 percent. The output voltage of the rate generator from pin E to pin D must be in phase ±2 degrees with the voltage from pin E to F with increasing bearing indications as determined by an approximate 16-volt measurement between pin D and F. With motor stalled and 18 volts applied to all windings, the rate-generator leakage measured between pins E and D shall not exceed 12 millivolts.

4.2.1.2.3 Tracking operation.- With 18 volts, 400 cps applied to phase one of the motor-rate generator drive circuit, the pointer shall start from any position and continue to rotate smoothly first clockwise and then counterclockwise when a 90-degree lagging and then a 90-degree leading 400-cps variable voltage applied across phase two is slowly increased from 0 to a maximum of 1 volt. This test is to be performed without tapping or vibrating the instrument.

4.2.1.2.4 Bearing measuring potentiometer accuracy.- The resistance of the card circuit of the potentiometer between pin k and m shall be so adjusted that the resistance as measured on a Wheatstone bridge is equal to 14,400 ohms ±1 percent. With the pointer servoed to 298 ±0.5 degrees as determined by the harmonic synchro resolver, the potentiometer shall be so set that the resistance across the contacts, pins n and p, is essentially a minimum, and so connected that a slight increase in bearing will cause a slight decrease in resistance between pins k and n, and between pins m and p. The angular accuracy of the sinusoidal potentiometer circuit shall be ±1.5 degrees when tested every 30 degrees with the exception of 90 and 270 degrees bearing by means of a push-pull sine wave voltage applied across the card, pins k and m.

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4.2.1.2.5 Bearing measuring resolver accuracy.- With the pointer set to 0 ±0.25 degree, the bearing measuring resolver shall be so set that when the rotor with a 390-ohm shunt resistor is fed from a 270-ohm source impedance, through a 560 ±1 percent ohm resistor, the phase shift between the input and the stator output with a 117,000 ±1 percent ohm resistor and a 0.01 ±1 percent microfarad capacitor connected in series will be 216 degrees. The 117,000-ohm resistor is to be connected to pin R and the 0.01 microfarad condenser to pin T. Pins P and S are to be grounded in this circuit. The common R and C output shall lead the 115 ±1 percent cps input voltage by 216 degrees with increasing lead resulting in decreasing pointer indications. The error in angular accuracy measured every 5 degrees for one revolution of the resolver, and every 30 degrees for one revolution of the pointer and determined by the resolver nulls, shall not exceed ±0.8 degree when tested by feeding the harmonic resolver output to a servo amplifier which will drive the indicator to the null positions of the harmonic resolver.

4.2.1.2.6 Differential synchro transmitter accuracy.- With the harmonic synchro resolver servoed to the 180-degree bearing, the differential shall be set to electrical zero. The absolute electrical zero position may be adjusted slightly to insure maintaining an accuracy of ±0.4 degree for the differential synchro with readings taken every 30 degrees in bearing.

4.2.1.2.7 Fundamental synchro resolver accuracy.- With the harmonic synchro resolver servoed to the 345-degree bearing, the fundamental resolver shall be set on electrical zero when the adjusting screw is set at its midrange of adjustment. The absolute electrical zero position may be adjusted slightly to insure maintaining an accuracy of ±0.5 degree for the fundamental resolver when tested for one revolution of the pointer with readings taken every 30 degrees in bearing.

4.2.1.2.8 Harmonic synchro resolver accuracy.- The error in angular accuracy measured every 5 degrees for one revolution of the resolver, and every 30 degrees for one revolution of the pointer as determined by the resolver nulls, shall not exceed ±0.5 degree on the pointer when read against the dial.

4.2.1.3 Environmental and mechanical tests.- Environmental and mechanical tests shall be conducted as specified below in accordance with Specification MIL-T-5422, except that under all conditions the indicator may be mounted to simulate operation on a vibration-insulated aircraft-type instrument board in accordance with Specification MIL-I-7028, or at the discretion of the contractor, the mechanical tests may be performed in accordance with Specification MIL-E-5272, as applicable to aircraft instruments.

4.2.1.3.1 Temperature altitude.- Temperature and altitude tests shall be conducted during which electrical tests shall be performed to determine that:

(a) Rotation speed meets the requirements of 4.2.1.2.1, except that the time allowed shall not exceed 120 seconds.

(b) Tracking operation meets the requirements of 4.2.1.2.3, except that the maximum voltage allowed shall not exceed 2 volts.

(c) The Bearing measuring potentiometer accuracy is maintained within the limits specified in 4.2.1.2.4 and the resistance does not exceed the specified limits by more than 1 percent.

(d) The Bearing measuring resolver accuracy is maintained within the limits specified in 4.2.1.2.5.

(e) The Harmonic synchro resolver accuracy is maintained within the limits specified in 4.2.1.2.8.

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4.2.1.3.2 Vibration.- Vibration tests shall be conducted during which Electrical tests shall be performed to determine that:

(a) Rotation speed meets the requirements of 4.2.1.2.1.

(b) There shall be no intermittent operation of the indicator circuits nor any evidence of electrical modulation caused by vibrating the instrument.

4.2.1.3.2.1 Stability.- After the Vibration tests, Electrical tests shall be conducted in accordance with 4.2.1.2.4, 4.2.1.2.5, 4.2.1.2.6, 4.2.1.2.7, and 4.2.1.2.8.

4.2.1.3.3 Shock.- Shock tests shall be conducted during which the indicator need not be operated. After the Shock tests, the instrument shall be tested and shall comply with the Electrical tests of 4.2.1.2.

4.2.2 Service approval tests.- The service approval sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the procuring activity.

4.2.2.1 Accessory materials and design data.- In addition to the complete equipment submitted, the contractor shall submit the accessory material and design data as specified in Specification MIL-E-5400.

4.2.3 Scope of tests.- Preproduction tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Preproduction tests shall include testing in accordance with Specification MIL-T-5422 and Specification MIL-I-6181. The design and test data supplied by the contractor shall indicate the equipments physical and electrical characteristics and establish that the equipment meets all the requirements of this specification. The test data shall include data obtained by the contractor in testing as specified in Specification MIL-T-5422.

4.2.4 Preproduction sample approval.- Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests.- Each equipment shall be examined carefully to determine compliance with the requirements of this specification with respect to material and workmanship. In addition, the equipment shall be given an operational check, with sufficient data recorded to assure satisfactory operation. As a minimum each equipment shall be tested to determine compliance with the following:

(a) Materials, dimensions, workmanship, marking, and finish comply with this specification.

(b) Operation and accuracy falls within the limits specified under Electrical tests of 4.2.1.2.

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4.3.2 Sampling tests.- Sampling tests shall be conducted by the contractor and shall include a performance check at ambient laboratory conditions. This test may also include any of the other Preproduction tests deemed necessary to determine conformance of the equipment with the requirements of this specification. These tests shall include also a careful examination of the way the equipment is packaged and packed to determine conformance with the packaging and packing requirements of this specification or the contract. Sampling tests shall be conducted on three indicators selected by the Inspector at random from each lot of 500 or less which have passed the Individual tests. The three indicators from each lot shall be submitted to and shall comply with the Environmental and mechanical tests of 4.2.1.3, except that only Step 2 and 6 of the "Test procedure" paragraph of the "Humidity test" of Specification MIL-T-5422 shall be applied under Temperature altitude tests of 4.2.1.3.1, and the Shock test of 4.2.1.3.3 of this specification shall not be required.

4.3.3 Rejection and retest.-

4.3.3.1 Indicators which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.3.3.2 When an indicator fails in the Sampling tests, the cause of failure shall be determined and corrected and the corrections shall be applied to the full lot from which the samples were taken. That portion of the Sampling tests in which failure occurred shall be repeated on three additional indicators.

4.4 Test procedures.- The test procedures and detail methods for conducting both Preproduction tests and Acceptance tests shall be prepared by the manufacturer and sent to the procuring activity for approval. The right is reserved by the procuring activity or the Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

5. PREPARATION FOR DELIVERY

5.1 General.- The indicator shall be preserved, packaged, and packed in accordance with Specification MIL-P-5633. Unit quantities shall be one each.

5.2 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified in the purchase document*
INDICATOR, AZIMUTH; ID-307/ARN
Specification MIL-I-8691A

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification as provided by the procuring activity.

6. NOTES

6.1 Intended use.- The azimuth indicator ID-307/ARN is an airborne bearing indicator suitable for use with an RT-220/ARN-21-type receiver-transmitter to provide the pilot of an equipped aircraft with meter indications of the bearing to a navigation beacon of known location.

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6.2 Ordering data.- Requisitions, contracts, and orders should state the conditions for the following.

6.2.1 Approval of Preproduction test sample.- It is expected that the contract or purchase order will specify that one equipment will be required as a preproduction sample and that this preproduction sample will be subjected to the Preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 When the requirements of the contract, this specification, or subsidiary specifications, publications, and drawings are in conflict, the contract should take precedence over all specifications, publications, and drawings, and this specification should take precedence over all other subsidiary specifications, publications, and drawings. Any deviation from this specification or from subsidiary specifications, publications, and drawings should be specifically approved in writing by the procuring activity.

6.4 Definitions and standards.-

6.4.1 Standard conditions.- Standard conditions refer to the following approximate conditions found in an ordinary laboratory:

(a) Ambient temperature: 20°C to 30°C (68°F to 86°F).

(b) Relative humidity: Not to exceed 90 percent.

(c) Pressure: Normal atmosphere.

(d) Vibration: As encountered in bench operation.

(e) Power: 26.5 volts dc, 115 volts, 400 cps ac.

(f) Warmup period: 10 minutes.

(g) Mounting: Mounted on instrument board

(h) Adjustments: All adjustments made in accordance with the manufacturer's instructions.

6.4.2 Precision resolver.- Precision resolver refers to a calibrated resolver mounted in a compass rose. The scale should be accurate to any angle within 0.1 degree, and the resolver should be capable of being set to any angle within 0.1 degree.

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